

ANSWER KEY — Integration Lesson 18 Test: Substitution II — u-Substitution

For the grader · full working shown · give credit for correct method with small arithmetic slips · missing "+ C" costs half a point

1. For $\int 2x(x^2+4)^6 dx$, state u and du .

Work: u is the inside of the power: $u = x^2+4$. Then $du = (\text{derivative of } u) dx = 2x dx$. Bonus observation: the $2x$ factor is present exactly, so u -substitution will work. **Answer: $u = x^2+4$, $du = 2x dx$.**

2. Compute $\int 2x(x^2-3)^4 dx$.

Work: (1) $u = x^2-3$. (2) $du = 2x dx$ — exact match. (3) Swap: $\int u^4 du$. (4) Integrate: $u^5/5$. (5) Back: $(x^2-3)^5/5 + C$. (6) Check: derivative = $5(x^2-3)^4 \cdot 2x / 5 = 2x(x^2-3)^4$ ✓. **Answer: $(x^2-3)^5/5 + C$.**

3. Compute $\int 3x^2(x^3+5)^3 dx$.

Work: $u = x^3+5$, $du = 3x^2 dx$ — exact match. Swap: $\int u^3 du = u^4/4$. Back: $(x^3+5)^4/4 + C$. Check: $4(x^3+5)^3 \cdot 3x^2 / 4 = 3x^2(x^3+5)^3$ ✓. **Answer: $(x^3+5)^4/4 + C$.**

4. Compute $\int \cos x (\sin x)^4 dx$.

Work: $u = \sin x$, $du = \cos x dx$ — exact match. Swap: $\int u^4 du = u^5/5$. Back: $(\sin x)^5/5 + C$. Check: $5(\sin x)^4 \cdot \cos x / 5 = \cos x (\sin x)^4$ ✓. **Answer: $(\sin x)^5/5 + C$.**

5. Compute $\int x(x^2+1)^3 dx$.

Work: $u = x^2+1$, $du = 2x dx$. We only have $x dx$ — off by the constant 2, so borrow: $x dx = du/2$. Swap: $\int u^3 \cdot (du/2) = 1/2 \cdot u^4/4 = u^4/8$. Back: $(x^2+1)^4/8 + C$. Check: $4(x^2+1)^3 \cdot 2x / 8 = x(x^2+1)^3$ ✓. **Answer: $(x^2+1)^4/8 + C$.** (Common error: forgetting the borrow and writing $(x^2+1)^4/4$.)

6. Compute $\int x^2(x^3-4)^3 dx$.

Work: $u = x^3-4$, $du = 3x^2 dx$, so $x^2 dx = du/3$. Swap: $\int u^3 \cdot (du/3) = 1/3 \cdot u^4/4 = u^4/12$. Back: $(x^3-4)^4/12 + C$. Check: $4(x^3-4)^3 \cdot 3x^2 / 12 = x^2(x^3-4)^3$ ✓. **Answer: $(x^3-4)^4/12 + C$.**

7. Does u -substitution work on $\int (x^3+2)^5 dx$?

Work: candidate $u = x^3+2$ needs $du = 3x^2 dx$, so the integral would need an x^2 factor (up to a constant). There is NO x factor of any kind, and only constants may be borrowed — you can never divide by $3x^2$. So u -substitution fails. The legal fallback: expand $(x^3+2)^5$ into a polynomial and integrate term by term with the power rule. **Answer: No — the derivative of the inside ($3x^2$) does not appear as a factor; expand and integrate term by term instead.**

8. Is $\int x(x^2+7)^2 dx = (x^2+7)^3/6 + C$ correct?

Work: differentiate the claim. d/dx of $(x^2+7)^3/6$: power rule brings down the 3 $\rightarrow 3(x^2+7)^2/6 = (x^2+7)^2/2$; chain rule multiplies by the inside's derivative $2x \rightarrow (x^2+7)^2 \cdot 2x / 2 = x(x^2+7)^2$. That matches the integrand exactly. **Answer: Yes, the classmate is right — the derivative gives back $x(x^2+7)^2$ ✓.**

9. Compute $\int_0^1 4x(x^2+1) dx$.

Work: $u = x^2+1$, $du = 2x dx$, so $4x dx = 2 du$. Indefinite: $\int u \cdot 2 du = 2 \cdot u^2/2 = u^2$. Back: $(x^2+1)^2$. Check: derivative = $2(x^2+1) \cdot 2x = 4x(x^2+1)$ ✓. FTC: at $x = 1$: $(1+1)^2 = 4$; at $x = 0$: $(0+1)^2 = 1$; subtract: $4 - 1 = 3$. **Answer: 3.**

10. Water flows in at $r(t) = 2t(t^2+1)$ L/min. How much flows in from $t = 0$ to $t = 2$?

Work: total = $\int_0^2 2t(t^2+1) dt$. $u = t^2+1$, $du = 2t dt$ — exact match. Indefinite: $\int u du = u^2/2 \rightarrow$ back: $(t^2+1)^2/2$. Check: derivative = $2(t^2+1) \cdot 2t / 2 = 2t(t^2+1)$ ✓. Plug in: at $t = 2$: $(4+1)^2/2 = 25/2$; at $t = 0$: $(0+1)^2/2 = 1/2$; subtract: $25/2 - 1/2 = 24/2 = 12$. **Answer: 12 liters.**